R104: cleared N51 leading residue at x=1

5 November 2026 | marked-23 Koblitz source-order experiment

R102 has 22 coefficients of order -9. Their leading term factors as
H = y^2 (y+1)^4 (l^2+l+1)^7. The fiber equations are
f = y^3(y+1)^4 and G = (l^2+l+1)^8.
Pinned Singular checks H is nonzero in the rank-112 source order.

y=0 component	y=1 component	reduced points
length 48; H=y^2 q^7	length 64; H=0	residues 0 and 0

Fresh Singular replay matched exactly. Independent F2 support and Lean checks verified the factorization and the component dimensions.
The least clearing exponent in this monogenic order is 9; normalized branch valuations still require its integral closure.
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